

Coding-Agent VLN — two-harness experiment report

claude-SDK path (beta-coding-agent) vs mini-swe-agent path (beta-react-harness)

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SETUP

Task / environment

R2R-CE (VLN-CE), split rand100 (the 100-episode paper-canonical subset of val_unseen). Simulator: habitat-sim via the env_habitat nodeset, hosted as an auto_host HTTP server (POST /call/{fn}); one server per parallel worker. Metrics come from habitat's own measures via env_habitat_evaluate (success = STOP within 3 m of goal; SPL; nDTW; oracle_success; steps).

Agent-facing surface (identical in both harnesses)

Tools: observe() -> egocentric RGB (+ clearance readout from depth in the left/center/right thirds), step(actions) with 0=STOP 1=FWD-0.25m 2=L15 3=R15 (budget broadcast + STOP confirmation gate), look_around() -> four labeled views (costs 24 turn steps). No pose, no map, no metrics exposed. System prompt embeds the instruction; identical text modulo one trailing newline (jinja convention). Tool schemas byte-identical (gated by beta-react-harness/check_equivalence.py).

Harness A: claude-SDK path (beta-coding-agent/)

Claude Agent SDK session per episode; tools served by an MCP stdio bridge (mcp_bridge.py) forwarding to the auto_host. SDK-side context management / thinking config apply. Subscription auth.

Harness B: mini-swe-agent path (beta-react-harness/)

mini-swe-agent 2.4.5 DefaultAgent per episode (pip library, loop unchanged); NodesetToolSet wraps the same auto_host as litellm tool-calls; multimodal tool results (image + status JSON). Linear message history, no compaction (image window off, i.e. full history sent; prefix caching on, default end). API-key billing via litellm. Limits: step_limit 100 LLM calls, env step budget 500, \$5/episode cost cap, 2400 s wall clock.

Conditions

All beta-react-harness runs: full mechanisms, no skill text.
SDK baselines as labeled: "bare" = observe/step only, no tuned mechanisms;
"mechanisms, no skill" = clearance + budget broadcast + STOP gate + look_around; "+<name>" = mechanisms plus that SKILL.md in the system prompt.

Known configuration difference

Turn budget differs between harnesses: SDK runs used max_turns = 80;
beta-react-harness runs used step_limit = 100 LLM calls. Episodes in which the mini agent used more than 80 LLM calls: sonnet 20/50 (7 of them ended successful), opus 11/50 (4 successful). The turn-budget broadcast shown to the agent tracked the respective limit in each harness.

Statistical note

Same episode indices across harnesses allow paired per-episode comparison; discordant-pair counts and exact McNemar p-values are reported as data.

beta-react-harness runs (full mechanisms, no skill)

run	model	eps	n	SR	SPL	nDTW	oracle	near-miss(3-5m)	stop	cost
smoke2	sonnet-5	0	1	0.00	0.00	0.74	0.00	1	1.00	\$0.18
calib10	sonnet-5	0-9	10	0.20	0.14	0.50	0.30	3	0.90	\$2.66
sonnet10_49	sonnet-5	10-49	40	0.50	0.35	0.45	0.55	3	0.95	\$14.41
opus10	opus-4-8	0-9	10	0.20	0.17	0.61	0.40	5	1.00	\$6.17
opus10_19	opus-4-8	10-19	10	0.80	0.59	0.59	0.90	0	1.00	\$7.84
opus20_49	opus-4-8	20-49	30	0.37	0.33	0.51	0.47	5	1.00	\$13.26
opus 0-49 combined		0-49	50	0.42	0.35	0.55	0.54	10	1.00	\$27.27
sonnet 0-49 combined		0-49	50	0.44	0.31	0.46	0.50	6	0.94	\$17.08

near-miss = failures with 3.0 m <= distance_to_goal <= 5.0 m at stop.

claude-SDK baselines (beta-coding-agent)

run	model	condition	eps	n	SR	SPL	nDTW	hear-mis	stop
sonnet100	sonnet-5	mechanisms, no skill	0-99	100	0.19	0.16	0.49	17	0.35
opus50_bare	opus-4-8	bare	0-49	50	0.48	0.46	0.59	5	0.72
opus50_ledger	opus-4-8	+ledger-nav	0-49	50	0.50	0.43	0.59	8	1.00
gen50_ledger	sonnet-5	+ledger-nav	0-49	50	0.44	0.30	0.46	5	1.00
fable50_bare	fable-5	bare	0-49	50	0.64	0.57	0.58	4	0.78
fable25_ledger	fable-5	+ledger-nav	0-24	25	0.68	0.50	0.57	4	1.00
fable25_49_ledger	fable-5	+ledger-nav	25-49	25	0.56	0.42	0.57	3	1.00
haiku50_ledger	haiku-4-5	+ledger-nav	0-49	50	0.20	0.12	0.19	3	0.78
calib10c	(default)	mechanisms, no skill	0-9	10	0.30	0.30	0.53	4	0.50
tune1_budget	sonnet-5	+ledger-nav (iter1)	0-9	10	0.10	0.10	0.45	3	1.00
tune2_pano	sonnet-5	+ledger-nav (iter2)	0-9	10	0.20	0.17	0.46	3	1.00
tune3_clearance	sonnet-5	+ledger-nav (iter3)	0-9	10	0.30	0.21	0.40	2	1.00
tune4_placement	sonnet-5	+ledger-nav (iter4)	0-9	10	0.30	0.22	0.45	3	1.00
tune5_stopgate	sonnet-5	+ledger-nav (iter5)	0-9	10	0.40	0.23	0.38	2	1.00
opus_skill_v1	opus-4-8	+opus-nav	0-9	10	0.30	0.22	0.48	3	1.00
opus_skill_v2	opus-4-8	+opus-nav	0-9	10	0.30	0.30	0.53	1	1.00
opus_skill_v3	opus-4-8	+opus-nav	0-9	10	0.10	0.10	0.55	4	0.20
opus_skill_v4	opus-4-8	+opus-nav	0-9	10	0.30	0.28	0.52	3	1.00

Same-episode-subset comparison

condition	n	SR	SPL	nDTW	oracle	near-miss
eps 0-9						
mini opus	10	0.20	0.17	0.61	0.40	5
SDK opus bare	10	0.30	0.28	0.44	0.40	0
SDK opus +ledger	10	0.20	0.20	0.52	0.40	5
mini sonnet	10	0.20	0.14	0.50	0.30	3
SDK sonnet no-skill	10	0.00	0.00	0.43	0.20	3
eps 10-19						
mini opus	10	0.80	0.59	0.59	0.90	0
SDK opus bare	10	0.60	0.57	0.69	0.80	0
SDK opus +ledger	10	0.50	0.43	0.61	0.70	1
mini sonnet	10	0.30	0.24	0.37	0.50	1
SDK sonnet no-skill	10	0.20	0.11	0.45	0.40	0
eps 20-49						
mini opus	30	0.37	0.33	0.51	0.47	5
SDK opus bare	30	0.50	0.48	0.60	0.57	5
SDK opus +ledger	30	0.60	0.51	0.60	0.60	2
mini sonnet	30	0.57	0.39	0.47	0.57	2
SDK sonnet no-skill	30	0.23	0.21	0.51	0.37	5
eps 0-49						
mini opus	50	0.42	0.35	0.55	0.54	10
SDK opus bare	50	0.48	0.46	0.59	0.58	5
SDK opus +ledger	50	0.50	0.43	0.59	0.58	8
mini sonnet	50	0.44	0.31	0.46	0.50	6
SDK sonnet no-skill	50	0.18	0.15	0.49	0.34	8

Paired comparison (counts)

PAIRED PER-EPIISODE SUCCESS COUNTS (identical episode indices)

mini opus vs SDK opus bare
n=50 both=14 first-only=7 second-only=10 neither=19
exact McNemar p=0.6291

mini opus vs SDK opus +ledger
n=50 both=14 first-only=7 second-only=11 neither=18
exact McNemar p=0.4807

mini sonnet vs SDK sonnet no-skill
n=50 both=7 first-only=15 second-only=2 neither=26
exact McNemar p=0.0023

Appendix – per-episode: mini opus 0-49

ep	succ	dtg(m)	nDTW	env steps	LLM calls	end	cost
0	0	4.5	0.66	92	25	stop_called	\$0.17
1	0	3.4	0.79	55	15	stop_called	\$0.09
2	0	3.4	0.78	130	52	stop_called	\$0.40
3	0	3.8	0.84	289	59	stop_called	\$0.52
4	0	14.4	0.15	441	96	stop_called	\$1.22
5	1	2.3	0.87	254	77	stop_called	\$0.74
6	0	13.7	0.31	354	81	stop_called	\$0.86
7	0	3.5	0.78	233	64	stop_called	\$0.53
8	0	5.2	0.30	327	86	stop_called	\$0.86
9	1	2.1	0.60	341	79	stop_called	\$0.79
10	1	2.1	0.77	253	79	stop_called	\$0.82
11	1	0.3	0.92	308	66	stop_called	\$0.64
12	1	0.9	0.96	113	23	stop_called	\$0.15
13	1	0.4	0.90	239	91	stop_called	\$0.90
14	0	16.8	0.00	391	86	stop_called	\$1.03
15	1	1.9	0.63	268	87	stop_called	\$0.88
16	1	1.1	0.91	112	26	stop_called	\$0.17
17	1	1.4	0.24	421	99	stop_called	\$1.14
18	1	0.5	0.50	421	96	stop_called	\$1.16
19	0	5.4	0.10	457	78	stop_called	\$0.96
20	0	3.7	0.64	422	99	stop_called	\$1.38
21	0	5.0	0.59	167	34	stop_called	\$0.28
22	0	8.9	0.53	251	85	stop_called	\$0.89
23	1	1.7	0.88	137	29	stop_called	\$0.21
24	0	8.3	0.37	156	39	stop_called	\$0.28
25	1	0.7	0.94	76	21	stop_called	\$0.12
26	0	14.8	0.03	250	57	stop_called	\$0.49
27	0	5.9	0.64	134	41	stop_called	\$0.34
28	1	0.8	0.91	148	49	stop_called	\$0.38
29	1	0.6	0.92	150	35	stop_called	\$0.25
30	1	1.9	0.27	200	43	stop_called	\$0.37
31	0	19.2	0.17	175	59	stop_called	\$0.49
32	1	0.6	0.95	136	46	stop_called	\$0.37
33	0	3.7	0.78	231	63	stop_called	\$0.59
34	0	5.7	0.38	332	79	stop_called	\$0.82
35	1	1.3	0.79	126	55	stop_called	\$0.39
36	0	4.1	0.68	103	24	stop_called	\$0.16
37	1	1.0	0.92	158	45	stop_called	\$0.34
38	1	2.6	0.41	194	65	stop_called	\$0.62
39	1	1.9	0.48	123	32	stop_called	\$0.26
40	0	19.5	0.08	117	30	stop_called	\$0.19
41	0	8.6	0.43	130	32	stop_called	\$0.22
42	0	9.2	0.28	346	82	stop_called	\$0.96
43	0	6.5	0.48	347	74	stop_called	\$0.80
44	0	9.3	0.34	302	51	stop_called	\$0.53
45	0	3.5	0.38	111	22	stop_called	\$0.17
46	0	15.7	0.09	73	14	stop_called	\$0.10
47	1	0.9	0.49	222	47	stop_called	\$0.39

Appendix – per-episode: mini sonnet 0-49

ep	succ	dtg(m)	nDTW	env steps	LLM calls	end	cost
0	0	4.0	0.65	179	32	stop_called	\$0.11
1	0	3.9	0.64	93	17	stop_called	\$0.06
2	0	8.0	0.59	173	45	stop_called	\$0.14
3	0	17.9	0.11	409	72	stop_called	\$0.32
4	0	7.1	0.33	495	81	stop_called	\$0.41
5	0	4.8	0.73	398	93	stop_called	\$0.49
6	0	15.8	0.18	275	47	stop_called	\$0.18
7	0	12.9	0.07	500	74	budget	\$0.54
8	1	0.4	0.80	235	43	stop_called	\$0.13
9	1	2.0	0.86	380	62	stop_called	\$0.29
10	0	17.6	0.02	345	86	stop_called	\$0.38
11	1	0.5	0.94	407	86	stop_called	\$0.47
12	0	5.0	0.07	465	96	stop_called	\$0.48
13	0	3.7	0.72	497	52	stop_called	\$0.34
14	0	13.5	0.00	374	92	stop_called	\$0.49
15	1	1.3	0.62	446	92	stop_called	\$0.42
16	1	1.9	0.84	89	27	stop_called	\$0.07
17	0	13.5	0.06	500	75	budget	\$0.39
18	0	5.8	0.39	493	74	stop_called	\$0.39
19	0	19.6	0.00	398	91	stop_called	\$0.56
20	1	1.0	0.89	387	62	stop_called	\$0.32
21	1	1.3	0.87	267	45	stop_called	\$0.21
22	0	8.9	0.55	386	92	stop_called	\$0.39
23	1	1.4	0.86	311	34	stop_called	\$0.16
24	0	8.8	0.32	458	89	stop_called	\$0.50
25	1	0.9	0.54	500	82	stop_called	\$0.41
26	1	0.2	0.04	493	91	stop_called	\$0.43
27	0	12.5	0.19	334	61	stop_called	\$0.25
28	1	2.3	0.62	494	78	stop_called	\$0.40
29	1	0.0	0.82	259	38	stop_called	\$0.13
30	1	1.8	0.27	223	41	stop_called	\$0.13
31	0	20.6	0.05	396	94	stop_called	\$0.47
32	1	0.4	0.66	487	79	stop_called	\$0.40
33	0	7.7	0.52	338	46	stop_called	\$0.33
34	0	8.8	0.18	479	77	stop_called	\$0.50
35	1	0.3	0.88	176	45	stop_called	\$0.12
36	1	1.3	0.83	336	67	stop_called	\$0.40
37	1	1.3	0.89	165	34	stop_called	\$0.09
38	1	2.2	0.81	269	48	stop_called	\$0.21
39	0	10.7	0.01	471	81	stop_called	\$0.38
40	0	33.2	0.00	401	96	stop_called	\$0.39
41	0	8.6	0.26	286	47	stop_called	\$0.21
42	1	2.1	0.85	303	82	stop_called	\$0.36
43	1	1.7	0.67	382	82	stop_called	\$0.32
44	0	4.4	0.39	496	78	stop_called	\$0.42
45	1	2.0	0.28	449	88	stop_called	\$0.64
46	0	33.1	0.00	480	92	stop_called	\$0.66
47	1	1.3	0.63	397	75	stop_called	\$0.33

Artifacts & reproduction

ARTIFACTS

Repository (branch dev/coding-agent):

```
beta-react-harness/      mini-swe-agent harness (agent/model/env/
                           toolset/driver + check_equivalence.py)
beta-coding-agent/       claude-SDK harness (bridge + driver + skills)
outputs/archive/react-harness-m1-2026-07-13/
                           mini trajectories: summary.json,
                           episode_{i}.jsonl, raw/ per run
outputs/archive/coding-agent-sdk-baselines-2026-07-13/
                           SDK baseline trajectories (19 runs)
```

Viewing logs:

```
AgentCanvas Coding-Agent Monitor -> Logs -> source toggle
[Claude SDK | mini-swe-agent] browses either path's runs (text log with
inline frames for local runs; archived runs have text logs only).
```

Reproducing a mini run:

```
1) start env servers (one per worker):
   PYTHONPATH=agentcanvas/backend:. ~/miniforge3/envs/ac-vlnce/bin/python \
   -m app.server.auto host --file workspace/nodesets/env/env_habitat.py \
   --class EnvHabitatNodeSet --port 9200
2) ANTHROPIC_API_KEY=... python beta-react-harness/run_episodes.py \
   --episodes 0-9 --split rand100 --model claude-sonnet-5 \
   --server-urls http://127.0.0.1:9200
```